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Chapter 4 Appendix A: A Scholarly Review of the Fiber Framework

> **Independent Publication Note**: This appendix is based on an independent scholarly paper and retains the self-reference "this paper" within the overall book structure. Other main text chapters uniformly use the self-reference "this chapter."

Abstract

The concept of the "fiber" has a long history in mathematics; Steenrod's (1951) fiber bundle theory laid its rigorous topological foundation. In mathematics, the fiber is the inverse image of a mapping; research in measurement theory, partial identification, information theory, and conformal prediction treats the relation between observation and state from different problem contexts, forming several research paths that admit structural comparison; fields such as AI evaluation have also seen works that use similar terms but with different definitions.

This paper systematically reviews the concept of "fiber" and related ideas in these traditions, identifies their core propositions and conceptual boundaries, and positions the original contribution of this paper's fiber framework within this academic landscape. The position advanced here is: the fiber framework does not invent the insight of "fiber as an epistemological tool," but establishes a dialogue among these parallel traditions, integrating the scattered insights of measurement theory, information theory, econometrics, and hermeneutic circle theory into an **integrative analytical framework** with "pseudo-small fiber" and "gradient of detectability" as core operational concepts, and explicitly directing it toward uncertainty management and cognitive debt accountability.

Keywords

Fiber; pseudo-small fiber; fiber bundle; partial identification; conformal prediction; information bottleneck; hermeneutic circle; epistemology; AI evaluation

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§1 Introduction: The Academic Genealogy of the Fiber Concept

"Fiber" is a precise concept in mathematics: given a mapping $f : X \to Y$, the fiber of y, $f^{-1}(y)$, is the set of all elements x satisfying $f(x) = y$. This concept received systematic development in topology—Steenrod's (1951) *The Topology of Fibre Bundles* established a rigorous formalized system for fiber bundles, revealing how the fiber bundle serves as a structure that is locally a direct product yet globally possibly nontrivial, describing the complex mapping relations between different spaces.

In recent decades, the mathematical fiber concept, methods with comparable structures, and works using similar terms with different definitions have developed in parallel across multiple disciplines. For a reading with a non-singleton inverse image under a given mapping, the observation cannot uniquely determine the true state; this structural problem must be distinguished from homonymous terminology. The philosophy of physics applies fiber bundle formalization to analyze the problem of observation depending on reference frames in general relativity (Bamonti, 2024). AI evaluation has seen the term "Fibristic Epistemology," which is homonymous with but different from the mathematical fiber (Bouattoura, 2025). Measurement theory establishes representational measurement theory, treating measurement as the construction of a homomorphism from an empirical relational structure to a numerical relational structure, rather than the direct reading of numbers from objects (Krantz et al., 1971). Econometrics develops the "partial identification" method, replacing the pretense of point identification with the reporting of identification regions (Manski, 2003). Information theory quantifies the change in uncertainty before and after observation (Shannon, 1948), and develops the Information Bottleneck method, using mutual information to measure the trade-off between compressing the input and retaining relevant information (Tishby et al., 1999). Machine learning develops "conformal prediction," generating prediction sets with a prespecified marginal coverage guarantee without specifying a parametric form for the data distribution, under applicable conditions such as exchangeability (Vovk et al., 2005; Angelopoulos & Bates, 2023). Sociology introduces the phenomenological concept of the "hermeneutic circle" into the analysis of social mechanisms, describing the epistemological circularity between partial information and holistic understanding (Zhao Dingxin, 2022).

These academic traditions have each developed similar structures within their own fields, using different terminological systems. The task of this paper is to systematically review these traditions, identify their core propositions and conceptual boundaries, and position the original contribution of this paper's fiber framework within this academic landscape.

Note on Reading Method. This review distinguishes three kinds of work: first, restating authors' claims in terms of their own concepts and argumentative scope; second, comparing structural connections between different theoretical objects while stating where they cannot be equated (e.g., comparing Manski's "identification region" structurally with the fiber); third, renaming and reorganizing the relevant questions in this book's terminology, which belongs to naming and classification within the framework and is annotated **[T]** (taxonomic

convention, see Chapter 4 §2.4). This annotation indicates only the provenance of the naming; it does not exempt the argumentation required for structural comparison and for subsuming concrete cases.

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§2 Mathematical Origins: The Rigorous Formalization of Fiber Bundles

§2.1 Steenrod (1951): The Topological Foundation of Fiber Bundles

Steenrod's (1951) *The Topology of Fibre Bundles* is the foundational work of modern fiber bundle theory. This book defines a fiber bundle as a mathematical object satisfying the following structure: a total space E, a base space B, a projection map $p : E \to B$, and a typical fiber F, such that for each point of B there exists a neighborhood U whose preimage $p^{-1}(U)$ is homeomorphic to the direct product $U \times F$ by a homeomorphism compatible with the projection ($p(\varphi(x, f)) = x$). This locally trivial yet potentially globally nontrivial structure makes fiber bundles a powerful tool for describing gauge field theory, general relativity, and other theories in modern physics. Steenrod's definition also includes a structure group G, which constrains the coordinate transformations between local trivializations; the diagnostic framework of this book does not use this additional structure. Steenrod uses the Möbius strip as the simplest example of a nontrivial fiber bundle: the base space is a circle and the fiber is a line segment; over a suitable open neighborhood in the circle, it is homeomorphic to the product of that neighborhood and the line segment, but globally it is not the trivial product of the circle and the line segment (Steenrod, 1951, §1.3).

From an epistemological perspective, two levels must be distinguished: the inverse image of a general mapping describes which elements produce the same reading—this is a set-theoretic triviality; fiber bundle theory further studies topological structures that are locally trivial yet possibly nontrivial globally. The basic diagnosis of this book relies mainly on the former, and the local-global relation provides only a supplementary analogy. **The tension between the local and the global**: a fiber bundle is locally a direct product (decomposable), but globally may possess twisting (not decomposable as a global direct product)—this locally simple yet globally complex structure provides a **structural analogy [T]** for understanding the following epistemological predicament: phenomena that appear simple in local observation may possess irreducible complexity at a global scale. It must be noted: this "structural analogy" is a retrospective appropriation by this paper of Steenrod's mathematical structure—Steenrod's original work is a pure mathematical construction and does not pursue epistemological applications. The fiber framework borrows this locally trivial yet globally nontrivial structure to analogize the local-global tension in epistemology; this is an illuminating analogy, but not a direct corollary of a mathematical theorem.

§2.2 From Mathematics to Epistemology: The Abstraction of the Concept

The spillover of the fiber concept from mathematics into epistemology essentially involves extracting its core intuition—"the same observation may correspond to multiple true states"—from rigorous mathematical structures and transforming it into an analytical tool applicable to any mapping scenario. This abstraction strips away the specific topological structures (homeomorphism, sections, structure groups, etc.) and retains the core operational definition: given a mapping function $\pi : \mathcal{T} \to \mathcal{O}$, the fiber $\pi^{-1}(r)$ of an observed value r is the set of all true states that could produce r. The size of the fiber can be measured in multiple ways—by cardinality, by measure, or, when the state space carries a probability distribution, by conditional entropy (the uncertainty

remaining after a given observation); these measures apply in different settings and do not necessarily induce the same ordering. Within the same state space, when a smaller fiber is a subset of a larger one, the observation rules out more possibilities.

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§3 Philosophy of Physics Tradition: Observation Depends on Reference Frames

§3.1 Bamonti (2024): Reference Frames and Physical Perspectivalism

Bamonti's (2024) preprint *Relational Observables and Physical Perspectivalism in General Relativity* represents a fiber bundle application of recent years in the philosophy of physics tradition. The paper uses fiber bundle formalization to address an interpretive question in general relativity: whether relational observables defined relative to different reference frames should be understood as partial characterizations of one and the same shared physical situation, or as physical situations that are comprehensive in their own right. To clarify the meaning of "perspective-neutrality," the paper distinguishes two properties: frame-independence (the preservation of values across changes of frame) and frame-freedom (construction without invoking any frame).

Bamonti's core move is to organize the space of models as a fiber bundle whose orbits are generated by the theory's symmetry group—the diffeomorphism group in general relativity—and to give, within a single construction, the gauge-equivalence classes, the relational observables defined relative to reference frames, and the maps between reference frames, so that two ontological interpretations can be formulated and compared within a single formal language. The values of physical quantities are defined relative to the physical system serving as the reference frame—the energy density, for instance, is constructed from the congruence of observers that constitutes the frame and depends irreducibly on which physical system serves as the frame; the relational observables of the same model can yield different gauge-invariant values under different reference frames. In the language of fiber bundles, selecting a reference frame amounts to introducing a smooth (local) section of the bundle, i.e., a gauge-fixing; the action of the symmetry group relates physically equivalent model representations within the same orbit. On top of this formalization, the paper puts forward two interpretations and does not adjudicate the choice between them: the "View from Nowhere" reads the frame-relative relational observables as partial characterizations of a shared physical situation—a situation represented by the gauge-equivalence class of whole models, corresponding to a qualified realization of moderate physical perspectivalism; the "View from Everywhere" reads what each relational observable characterizes as a physical situation that is comprehensive in its own right, without commitment to any shared frame-free reality, the gauge-equivalence class retaining only formal status, corresponding to a qualified realization of strong physical perspectivalism.

§3.2 Comparison with This Paper's Fiber Framework

Bamonti's work resonates with this paper's fiber framework at the level of formal tools, but the mathematical objects cannot be identified directly: in his fiber-bundle formalization, selecting a reference frame corresponds to a section σ, the bundle projection sending models to equivalence classes is defined separately, and frame selection can also induce a map onto the image of a section. This paper's π maps true states to observed readings; a structural analogy requires these correspondences to be specified—both use the fiber concept to describe the core insight that "observation depends on the mapping process," and in Bamonti's framework, the "observed value" is the reading on a specific section. A further distinction is required between the multiplicity

of model representations and the multiplicity of physical possibilities: the multiple models within a gauge orbit can be equivalent representations of the same physical state, and the multiple elements of a bundle fiber do not automatically imply uncertainty about the physical truth.

Yet fundamental differences exist between the two. **Domain of application**: Bamonti deals with gauge invariance problems in physics, where the sections, the bundle projection, the construction of reference frames, and the gauge transformations all have rigorous mathematical definitions (the diffeomorphism group, gauge transformations); this paper's fiber framework deals with general mapping scenarios such as AI systems and social cognition, where the mapping functions typically lack rigorous formalized expressions. **Guiding question**: Bamonti asks "how frame-related relational observables should be constructed and understood, and whether they should be interpreted as partial characterizations of one and the same shared physical situation or as physical situations that are comprehensive in their own right"; this paper's fiber framework asks "under conditions of an unknown mapping function, how can fibers be measured, shrunk, and annotated so as to manage uncertainty and track cognitive debt?" **Methodological positioning**: Bamonti uses fiber bundles as a rigorous mathematical tool; this paper's fiber framework uses fiber as a heuristic epistemological metaphor and analytical framework.

Bamonti's work provides a comparative case of using fiber bundle formalization to treat the problem of reference frames. This paper's fiber framework directly borrows the inverse-image concept of general mappings to analyze diagnostic problems different from Bamonti's objects of study—the two are parallel applications of the same mathematical intuition in different problem domains, not a relationship of inheritance. It must be noted: Bamonti keeps the choice between the two interpretations open; among them, the "View from Everywhere" does not commit to a shared frame-free reality, which differs from this book's realist presupposition of retaining the distinction between "true state" and "observed reading"; the comparison here is confined to the level of mathematical tools.

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§4 A Kindred Term in AI Evaluation: fibristic epistemology

§4.1 Bouattoura (2025): Fibristic Epistemology for AI Tool Evaluation

Bouattoura's (2025) paper *A Fibristic Epistemology for AI Tool Evaluation: A Logic-Based Simulation Framework* proposes the term "fibristic epistemology." It must be noted: that paper was published in *International Journal of Enhanced Research in Educational Development* (IJERED).[1] That paper is cited here **not as academic authority support**, but as a case of homonymous use of the term "fibristic" in AI evaluation; the questions it can be compared with this paper on are transparency and traceability, not a shared mathematical definition of the inverse image—similar diagnostic intuitions are growing independently. Bouattoura explicitly uses the term "fibristic epistemology," and its core criticism is that the existing narratives in the AI evaluation literature "frequently overestimate capacities, conveying alternative realities rather than empirical truths."

Bouattoura's framework is an evaluation system based on logic-weighted simulation and rule-based scoring. Its concrete implementation is 114 simulated user entries: each entry is composed of preset categorical attributes (gender, age, education, field of study, brainwave type, stress level, cognitive modifier, learning type), with values drawn from hard-coded distribution tables; evaluation proceeds by executing structured bonuses and

[1] This paper cites Bouattoura (2025) to compare a homonymous use of "fibristic" in AI evaluation and the paper's requirements for transparency and traceability, not to use the journal's standing as an authoritative basis for this book's framework.

penalties on preset base scores, with the sources of the bonuses and the final modifier factors item-by-item traceable in the output—the whole process is audit-traceable.

§4.2 Comparison with This Paper's Fiber Framework

Bouattoura's work resonates with this paper's fiber framework in its central concern: both criticize existing evaluation narratives for overestimating capacities, and both demand that the evaluation process be transparent and traceable. The term "fibristic" refers to a biological-structural metaphor of layering—research is described as a layered entity comprising a skeleton (invariants), muscles (flexibility), and ligaments (hyperbolic connections)—which is not the same mathematical object as the fiber of this paper, defined as the inverse image of a mapping.

Yet significant differences exist between the two in **methodological positioning** and **scope of application**. **Methodological positioning**: Bouattoura's framework is essentially a **methodology** for AI evaluation—it tells researchers "how AI systems should be evaluated"; its core contribution is a set of evaluation procedures based on logic-weighted simulation and rule-based scoring. This paper's fiber framework is essentially a **diagnostic theory**—it tells analysts "what structural problems exist in any mapping system"; its core contribution is conceptual tools such as pseudo-small fiber and gradient of detectability. This book further decomposes the fiber into analytical levels such as visible/hidden dimensions and the perceived fiber. **Scope of application**: Bouattoura's framework is specifically designed for AI tool evaluation; its formal tools (rule engines and preset distribution tables) are bound to simulated evaluation scenarios. The basic representation of this paper's fiber framework can be used across domains—from AI to financial markets to social indicators; whether it yields effective diagnosis depends on the task, the scope of states, the constraints on the mapping, and the calibration evidence. Cross-domain applicability is a design goal, not a guarantee of equal explanatory power for all observation problems.

Since the two share the word "fiber" but refer to different things—one a biological-structural metaphor with a rule-based evaluation procedure, the other the inverse image of a mapping—Bouattoura's work should not be viewed as a methodological application or theoretical precursor of this paper's framework. This review cites it as an example of the independent diffusion of the word "fiber" in evaluation contexts, and compares the transparency and traceability demands that the two share.

It is worth noting that Bouattoura's rule-based scoring provides an operational tool in concrete evaluation scenarios, whereas this paper's fiber concept encompasses information loss in any mapping process—including scenarios that cannot be formalized as rule-based evaluation (such as perceptual biases, institutional information filtering). The two do their work on different levels: "making the evaluation process traceable" and "making mapping losses visible."

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§5 Measurement Theory and Econometrics Traditions

§5.1 Krantz et al. (1971): Representational Measurement Theory

Krantz et al.'s (1971) *Foundations of Measurement* established the axiomatic foundation of modern measurement theory. Its core proposition is that measurement can be understood as the construction of homomorphisms

from empirical relational structures to numerical relational structures, while foundational analysis clarifies the assumptions required for such constructions. This paper accordingly links measurement representation to the problem of instrument calibration: the interpretation of numerical values requires specifying which empirical relations the mapping preserves and under what conditions; this is a retrospective rereading by this paper, not a reduction of the original work to a pseudo-small fiber theory **[T]**.

This proposition can be structurally compared with Proposition 1 (the existence of non-singleton fibers) and Proposition 2 (calibration narrows the identification set) of the fiber framework once the representational mapping is distinguished from the observation model of the instrument. When the actual mapping is fully known, the knowledge-conditioned identification set can reduce to that mapping's actual inverse image at the given reading; learning has not changed the actual inverse image but has excluded other candidate mappings. Insufficient knowledge of the mapping may enlarge the identification set; measurement error must be specified separately in the relevant observation model. This paper accordingly emphasizes: **the reliability of measurement cannot be guaranteed by the number of digits in the reading alone; it also depends on the calibration relationship between the mapping function and the truth**. This relational-structure perspective provides the basis for this paper's analysis of the interpretive conditions of readings; using it further to distinguish fiber size from pseudo-small fiber is this paper's diagnostic translation **[T]**.

§5.2 Manski (2003): Partial Identification

Manski's (2003) *Partial Identification of Probability Distributions* represents the most systematic response in econometrics to the predicament that "observation cannot uniquely determine the truth." Its core method is: when the population distribution of the observed variables and the adopted assumptions cannot uniquely determine the target parameter, report the identification region of the parameter compatible with these conditions, rather than passing off partial identification as point identification; under finite samples, a further distinction must be drawn between the identification region itself and its estimation and statistical uncertainty.

In the language of fibers, Manski's "identification region" can be compared structurally with the fiber: it is the set of target parameter values compatible with the observed distribution under a given statistical model and assumptions. Manski's "worst-case bounds" are obtained under explicit observational, sampling, and model specifications without adding certain stronger identifying restrictions; they are not conclusions derived from data alone without any assumptions. This is directionally consonant with the operation of "annotating the fiber" advocated by the fiber framework—not pretending the fiber is small, but honestly annotating the size of the fiber.

The core resonance between Manski's work and the fiber framework lies in the fact that both advocate **not passing off partial identification as point identification, and not masking identification uncertainty with a single number**. From the perspective of the fiber framework, if the evidence permits only partial identification, yet only "the parameter value is 3.7" is reported and the identification uncertainty is masked, a pseudo-small fiber may be manufactured—making the user believe the uncertainty is small, when in fact the fiber may be very large; the problem lies not in the form of point estimation itself, but in whether the report exaggerates the certainty of the evidence. An identification region with explicit assumptions stated ("the parameter value is between 2.1 and 5.3"), under the given assumptions, is equivalent to honestly annotating the size of the fiber. It must be noted: "pseudo-small fiber" is a conceptual tool proposed by this paper; Manski's original text does not use this term—this is a **retrospective rereading** of his methodology by this paper's framework **[T]**. The

fiber framework's path of "annotating the fiber" resonates operationally with Manski's practice of explicitly reporting the identification range and its assumptions, but a general observation inverse image, a parameter identification region, and a finite-sample interval cannot be directly identified with one another. Going further, Manski's position that the identification region should be reported when the evidence permits only partial identification, and his demonstration of the test use of comparing point estimates with identification region estimates, together embody a stance of accountability toward the knowledge consumer—a stance that already constitutes an econometric precursor to the auditing orientation of "annotating the fiber."

§5.3 The Connection to Auditing and Cross-Period Accountability

Measurement theory and partial identification theory provide profound methodological references for the fiber framework, and they also contain attention to assumption transparency and inferential credibility. The working focus of this paper is to link these issues with preference optimization, pseudo-small fibers, and the cross-period accountability of cognitive debt, not to claim that auditing concerns first appeared with this paper. On this foundation, the fiber framework advances a step further: it asks not only "how to shrink the fiber," but also "what mechanisms manufacture pseudo-small fibers" (such as the preference optimization of Reinforcement Learning from Human Feedback (RLHF)), "how does the accumulation of unmanifested consequences associated with fibers transmit through the system" (cognitive debt dynamics), and "who bears the consequences of fiber inflation" (auditing and accountability). The difference lies in the organization of the problems and the focus of application, not in a claim that existing theories lack auditing or accountability concerns.

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§6 Information Theory and Machine Learning Traditions

§6.1 Shannon (1948): Information Entropy

Shannon's (1948) information theory provides a quantitative language for the fiber framework. Information entropy $H(X)$ measures the degree of ignorance about X before observation; for discrete random variables, given the joint distribution, $H(X \mid Y = y)$ describes the probabilistic uncertainty remaining under a particular reading, and $H(X \mid Y)$ is the conditional entropy averaged over readings; they are not synonyms for the cardinality of the inverse image. Mutual information $I(X;Y) = H(X) - H(X|Y)$ quantifies how much the observation Y has reduced the uncertainty about X. The data processing inequality $I(X;Z) \leq I(X;Y)$ (for a Markov chain $X \to Y \to Z$) provides an information-theoretic formulation of Proposition 3: without introducing additional information, subsequent processing cannot increase the mutual information about X; the inequality allows equality and does not by itself prove that any given compression step is strictly lossy.

§6.2 Tishby et al. (1999): The Information Bottleneck Method

The Information Bottleneck method proposed by Tishby et al. (1999) is the development within the information theory tradition with the most structural resonance with the fiber framework. The core proposition of this method is: given an input variable X and a target variable Y, under a given tradeoff between compression and relevant-information retention, the optimal intermediate representation T retains the information relevant to the target Y while compressing its mutual information with the input X—i.e., retaining information relevant to the output and discarding the rest; this does not require each of the two objectives to attain its unconditional

optimum. Shwartz-Ziv and Tishby (2017) further argued on experimental grounds that the training process of deep neural networks can be described in two phases—first fitting ($I(T;Y)$ increases), then compression ($I(T;X)$ decreases). Subsequently, Saxe et al. (2019) showed through analysis and experiments that, under the deterministic-network and information-estimation settings they studied, particularly when mutual information was computed using simple binning, the two-phase compression phenomenon depended on the activation function: it could occur with double-sided saturating functions, whereas settings using ReLU did not exhibit the same trajectory; these results do not support treating the compression phase or its causal connection with generalization as a universal law of deep networks. This paper's reference to the Information Bottleneck method is limited to its conceptual contribution as an analytical framework (that the optimal intermediate representation should retain task-relevant information while discarding the rest), and does not depend on this framework's specific empirical characterization of the training stages of deep neural networks.

The Information Bottleneck method and the fiber framework share the data processing inequality as their mathematical foundation—both hold that once information is lost in a mapping, subsequent processing cannot recover it. Yet three key differences exist between the two, meaning the fiber framework is not an application or repackaging of Information Bottleneck theory.

Difference 1: Application Focus. The Information Bottleneck method studies task-relevant information compression in general and is not limited to inter-layer information flow within a single model; deep networks are one application. This paper's fiber framework places its analytical focus on the transmission of information—from true states through sensors, models, and user interfaces to user decisions—in multi-actor socio-technical systems, together with the associated institutional and cognitive conditions. This is a difference in application focus, not a principled restriction of the Information Bottleneck method to model internals.

Difference 2: Value Judgment. From the perspective of the fiber framework, the core operation of the Information Bottleneck—"discarding information irrelevant to the output"—is itself an action that needs auditing. The Information Bottleneck specifies, through the task variable and its joint distribution, which information is relevant to the goal, and analyzes the tradeoff between retaining relevant information and compressing the input—under appropriate conditions, discarding task-irrelevant information may aid generalization; this formalization does not replace an institutional audit of the choice of task itself. Dimensions deemed "irrelevant" and thereby discarded may contain tail risks, the interests of minority groups, and stability across institutional cycles. The arbiter of "irrelevant information" is the designer of the loss function, not all stakeholders in the true state. The Information Bottleneck has already discussed what should be retained for a given task; the fiber framework further presses who chooses the task, and which interests and consequences are left uncovered by that task.

Difference 3: Computability. The Information Bottleneck characterizes compression using mutual information $I(X;T)$ defined on a specified probabilistic model; its estimation still depends on data, modeling assumptions, and estimation methods—for continuous deterministic representations, the mutual information may be infinite; after binning, discretization, or noise injection, one must specify which variables and which model have been changed, and the resulting finite value cannot be unconditionally interpreted as the mutual information of the original representation. The fiber framework measures uncertainty using the "size" of the fiber $\pi^{-1}(r)$; in many open socio-technical settings it lacks sufficient information about the state space, mapping, and measure to calculate fiber size directly at present—this is a limitation of applicability and operationalization, and also the direction that the proxy indicator construction plan in Chapter 13 §8 attempts to remedy; when analysis extends to socio-technical systems without sufficient observations and constraints, precise measurement is often more

difficult, and the extent of this difficulty depends on the specific object and evidence conditions rather than on the category "socio-technical system" itself.

The Information Bottleneck provides a quantitative tool for analyzing task-relevant compression; this paper's fiber framework links these issues with the institutional diagnosis of pseudo-small fibers, pseudo-calibration, and cognitive debt. This linkage does not presuppose restricting the Information Bottleneck to model internals, nor does it claim that its mathematical tools cannot be used for multi-actor systems.

Although the fiber framework and the Information Bottleneck differ in application focus, the latter's core insight—optimal compression and task relevance—can provide an important dimension of refinement for the fiber framework. The objective function of the Information Bottleneck reveals: which information a mapping process discards and which it retains depends on the **task variable** Y it is optimized to predict. In the language of the fiber framework, this means that a system's hidden subspace $\mathcal{T}_h$ is not uniform—the information it contains includes some that is indeed irrelevant to the current task (under appropriate conditions, discarding it may aid generalization) and some that is crucial for an unoptimized task Y' (discarding it creates hidden risks). Therefore, a more refined diagnosis should not stop at "all mappings are lossy, so the hidden subspace is large," but should press further: "For which task Y was this mapping process optimized to preserve information? For other possible tasks, what potentially crucial information did it discard?" This line of questioning pushes the fiber framework's diagnosis from the generalized level of "all mappings are lossy" toward the refined level of "the sufficiency of a specific mapping for a specific set of tasks."

§6.3 Vovk et al. (2005) and Angelopoulos & Bates (2023): Conformal Prediction

Conformal prediction is a direct engineering interface in the field of machine learning for the problem of "annotating the fiber." Vovk et al.'s (2005) *Algorithmic Learning in a Random World* laid the theoretical foundation of conformal prediction; Angelopoulos and Bates's (2023) *Conformal Prediction: A Gentle Introduction* provided a systematic engineering implementation guide.

The core promise of conformal prediction is: under the assumption of data exchangeability, for any pre-trained model, a prediction set can be generated whose coverage rate has a probabilistic guarantee (e.g., in repeated sampling, the average coverage rate is no lower than a specified level). In the language of fibers, conformal prediction can be retrospectively reread as an engineering interface for uncertainty annotation, but its prediction sets are generally not the complete inverse image of a fixed observation mapping. Set size must be interpreted at a specified coverage level and under specified scoring, calibration, and task conditions; narrower sets may indicate greater predictive efficiency, but do not automatically establish smaller true fibers or conditional coverage for every input.

The core resonance between conformal prediction and the fiber framework lies in the fact that both advocate **replacing "point estimates" with "sets."** Reporting only a single answer while masking evidential uncertainty may manufacture a pseudo-small fiber; a prediction set with clear coverage conditions can express part of the predictive uncertainty. A point output is not in itself a pseudo-small fiber, and an uncalibrated set output is not automatically honest either. Conformal prediction is a borrowable method with explicit coverage guarantees on the fiber framework's "annotating the fiber" path.

This path is extending into the domain of large language models. Quach et al. (2024) demonstrated at ICLR the specific application of conformal prediction in generative language models: generating prediction sets that, with

a prescribed probability, contain at least one response meeting the set acceptance criteria, rather than a single point output. Campos et al. (2024), in a survey published in *Transactions of the Association for Computational Linguistics*, systematically reviewed the current application status and open problems of conformal prediction across various natural language processing tasks. These developments indicate that the engineering route of "annotating the fiber rather than pretending the fiber does not exist" is extending from traditional prediction tasks to generative AI—connecting with this paper's diagnosis of pseudo-calibration.

Claim-Type Annotation: Conformal prediction is a borrowable method with explicit coverage guarantees on the fiber framework's "annotating the fiber" path, but the two are not equivalent. What conformal prediction provides is a **coverage guarantee under a given model, provided the data satisfy an exchangeability condition**—it answers "how prediction sets are constructed under a given coverage level and exchangeability conditions, and what guarantee the average coverage attains." What the fiber framework presses is the **information loss structure at the mapping-chain level**—it answers "from the true state to the final reading, which dimensions are discarded at which layer." This paper retrospectively rereads the former as an engineering interface of the latter in a single-layer prediction task **[T]**, but it does not cover the multi-layer mapping chain accumulation effects, the institutional causes of pseudo-calibration, and the dynamic transmission of cognitive debt that the fiber framework is concerned with. Equating conformal prediction with the fiber framework confuses the scope of an engineering interface with that of a diagnostic framework; whether this further constitutes a pseudo-small fiber requires specifying the observation, the true uncertainty, and the observer's perception of it.

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§7 Sociology Tradition: Hermeneutic Circle Theory

§7.1 Zhao Dingxin (2022): The Hermeneutic Circle

Zhao Dingxin's (2022) "Levels of Explanation and the Hermeneutic Circle: Interpreting Social Mechanisms" introduces the concept of the "hermeneutic circle" into sociological analysis. Its core proposition is: social scientists can only ever start from partial information and attempt to understand the whole; the resulting theory is forever confined within a "hermeneutic circle," unable to fully escape the researcher's initial presuppositions and limited information. The core task of research is to "shrink the hermeneutic circle"—through five methods: the mutual corroboration of macro-structures and micro-mechanisms; the mutual corroboration of macro-structures, micro-mechanisms, and temporal sequences; the exclusion of alternative explanations; counterfactual reasoning; and increasing the amount of information about the problem being explained—but one can never "close the hermeneutic circle."

In the language of fibers, the size of the hermeneutic circle corresponds to the size of the fiber; the "hermeneutic circle going to zero" (author's translation)—which Zhao illustrates with the "true believer," who takes what he believes to explain everything—corresponds to the pseudo-small fiber (reading the zero as the false certainty of theory holders is a retrospective rereading by this paper **[T]**). Both share the same diagnosis: consistency is not calibration; confidence is not correctness.

§7.2 Resonance and Divergence with the Fiber Framework

The structural isomorphism between the two has been detailed in Chapter 4 §5. One additional key divergence should be added here: the hermeneutic circle theory is a strategy centered on mutual corroboration, excluding

competing explanations, and increasing the amount of information; the fiber framework is a strategy of "managing the medium"—auditing the mapping process itself, annotating uncertainty, introducing friction. The two strategies are not mutually exclusive but complementary: increasing the amount of information is necessary, but when the mapping process of information itself is systematically distorted, relying solely on increasing information may be counterproductive—repeatedly encountered biased information may keep reinforcing false certainty without adding discernment of the true state; whether it enlarges the identification set or aggravates the underestimation of uncertainty must be analyzed separately.

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§8 The Academic Positioning of This Paper's Fiber Framework

Within the above academic landscape, the originality of this paper's fiber framework lies not in inventing the insight of "fiber as an epistemological tool"—multiple academic traditions have independently developed similar ideas. The originality of this paper lies in integration across the following six dimensions:

First, integration of conceptual ecologies. Existing traditions each use different terminological systems (identification region, hermeneutic circle, conditional entropy, prediction set), with fewer direct cross-references among them. This paper constructs a unified conceptual ecology with "fiber" as the core metaphor—large fiber, small fiber, pseudo-small fiber, gradient of detectability, perceived fiber—making it convenient to compare these views and their conditions of applicability within a single language. This integration is not itself a new discovery, but a new organization—it does not add to the empirical content of existing traditions, but changes the relationships among them.

Second, the refinement of diagnostic concepts. Existing traditions are mainly concerned with "how to measure uncertainty" (measurement theory, partial identification) or "how to evaluate systems" (Bouattoura's fibristic epistemology); some traditions already contain implicit diagnostic intuitions—Manski's position of reporting identification regions under partial identification is itself a critique of "false certainty," and the Information Bottleneck's "discarding irrelevant information" also involves judgment about the cost of compression. On this basis, this paper makes this diagnostic intuition explicit by naming it "pseudo-calibration": when preference rewards systematically favor the appearance of certainty but cannot distinguish grounded from ungrounded certainty, and when independent calibration constraints are lacking, the system may learn not to reduce true uncertainty but to manufacture pseudo-small fibers. On the basis of the selected literature, this paper organizes this risk into the diagnostic vocabulary of "pseudo-small fibers / pseudo-calibration"; this naming, and the analysis of its formation mechanism (Chapter 4 §2.4), is part of this paper's analytical design and does not assert that the relevant mechanisms have never been studied in prior literature; its core intuition already has precursors in the engineering practices of partial identification and conformal prediction.

Third, the attempt at cross-domain universality. Existing traditions are mostly bound to specific domains (gauge invariance in physics, methodology of AI evaluation). The basic representation of this paper's fiber framework can be used across domains—from AI to financial markets to social indicators; whether it yields effective diagnosis depends on the task, the scope of states, the constraints on the mapping, and the calibration evidence. The complementary analysis with the Macro-Cognitive Accounting Framework (MCAF) in Chapter 4 §4, and the dialogue with the hermeneutic circle theory in §5, are both examples of this cross-domain use. But this positioning itself is not an empirical contribution—it is a design choice of the framework. The promise of

universality needs to be fulfilled through the scenario mappings of the subsequent chapters, not self-proclaimed in a review.

Fourth, the introduction of the auditing perspective. Existing traditions not only measure uncertainty, identify parameter sets, and evaluate systems, but also discuss assumption transparency, inferential credibility, traceability, and accountability; this paper's distinction lies in organizing these concerns together with pseudo-small fibers and the cross-period tracking of cognitive debt. This paper's fiber framework explicitly incorporates "accountability" into the framework design: it asks not only "how large is the fiber," but also "who is manufacturing pseudo-small fibers," "who bears the consequences of fiber inflation," and "how to establish a bookkeeping structure to track the accumulation of unmanifested consequences associated with fibers." This is an organizational and applied attempt by this paper, from uncertainty analysis toward cross-domain auditing uses (this "auditing perspective" builds on related concerns in prior research rather than inventing them for the first time; see the end of §5.2 for the connection with Manski).

Fifth, the complementary design with the MCAF framework. This paper further pairs the fiber framework with MCAF: the fiber framework provides micro-structural analysis (information loss and uncertainty distribution in a single mapping), and the MCAF framework provides macro-dynamic tracking (the accumulation, concealment, and liquidation of cognitive debt). The two are complementary and cannot substitute for each other.

Sixth, naming audit as a first-layer practice of pseudo-calibration diagnosis. This paper incorporates "naming" into pseudo-calibration diagnosis and develops the concrete practice in Chapter 13 and Appendix A—applying diagnostic logic to popular terms such as "Artificial Intelligence," "Large Language Model," and "token," arguing that the high-level associative words carried by these names substitute for precise but unglamorous structural descriptions, and proposing an alternative naming system—Reactive Lossy Mapping (RLM), Statistical Language Projector (SLP), Mapping Element (ME), etc. This practice is a concrete demonstration of the framework's movement from "diagnostic language" to "diagnostic operation": each use of RLM rather than AI annotates a diagnostic claim—this is not intelligence, but a reactive, lossy, uncalibrated mapping device with no temporal depth; whether these working names are more precise than the popular ones depends on whether the diagnostic dimensions they mark stand up to scrutiny, not on the act of renaming itself. McDermott (1976) criticized "wishful mnemonics," proposing either neutral placeholders like "G0034" or names revealing intrinsic mechanisms (such as NODE-NET-INTERSECTION-FINDER) in place of grand labels like "UNDERSTAND," and gave multiple contrastive examples. McDermott has already proposed concrete naming criticism and corrective methods. The further work of this paper is to incorporate this issue into the book's pseudo-calibration diagnosis, specifying name comparison, mechanism explanation, and conditions of use; what is claimed here is the book's internal systematization, not characterizing existing research as "isolated appeals," nor claiming to have proven the historical priority of this practice.

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§9 Conclusion: Positioning on the Shoulders of Giants

This paper's fiber framework is not an invention starting from zero, but an integration, deepening, and cross-domain expansion built upon multiple academic traditions. The paper's mathematical and methodological references include Steenrod's fiber-bundle mathematics, representational measurement theory, partial identification,

information theory, conformal prediction, and hermeneutic circle theory; Bamonti provides a parallel formalization in the philosophy of physics, while Bouattoura provides a comparison case that uses a different meaning of the term but likewise emphasizes transparency in evaluation.

The original contribution of this paper lies not in being "the first to propose fiber as an epistemological tool"—this insight has had multiple independent discoveries. The contribution of this paper lies in **constructing a unified analytical tool oriented toward uncertainty management and cognitive debt accountability, with "pseudo-small fiber" and "gradient of detectability" as core operational concepts; proposing a design for cross-domain application; and demonstrating several possible paths of conceptual collaboration and operationalization.**

Clarifying this academic positioning does not weaken the originality of this paper, but makes its originality more precise. Standing on the shoulders of giants is not to claim to be taller than the giants, but to see clearly what each giant saw, what each overlooked, and how to integrate their insights into a more complete analytical system.

Selectivity Annotation of the Review: Any review is selective. This review has chosen the academic traditions most directly structurally relevant to the fiber framework, but has omitted several potentially relevant directions: the sensitivity analysis of posterior distributions to prior choice in Bayesian epistemology (robust Bayesian analysis), discussions of ambiguity and Knightian uncertainty in decision theory, the classic debate on the underdetermination of theory in the philosophy of science, as well as the distinction between ensemble averages and time averages in Peters's (2019) ergodicity economics—the latter shares with this paper's "pseudo-small fiber" concept the intuition that "good performance on a spatial dimension cannot guarantee safety on a temporal dimension" (the "spatial/temporal dimension" phrasing is this book's translation; Peters himself formulates the distinction as expectation values versus time averages), but uses different conceptual tools, constituting an indirect resonance rather than a direct structural isomorphism. These directions have not been incorporated into the present comparison; hence this paper's judgments on originality and cross-domain applicability are limited to the selected literature. The basic definitions and their conditional corollaries can be examined on their own, but a more complete scholarly positioning requires an expanded scope of comparison.

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